Inflation Risk Assessment using Monte Carlo Simulation with Block Bootstrap

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Introduction

In this project, we use Monte Carlo Simulation to assess the risk that inflation poses to personal savings in the United States of America. The objective is to understand how much purchasing power 10,000 Dollars ($) will lose after 10 years, based on the historical behavior of American inflation.

To make this analysis as realistic as possible, we will not use a theoretical distribution (such as the normal distribution), but will instead apply **Moving Block Bootstrap**. This technique randomly samples consecutive blocks of years from history, thereby preserving time dependence and actual economic crises (fat tails).

Data Description

The data is obtained from the World Bank, indicator `FP.CPI.TOTL.ZG`, which measures the annual inflation rate. To guarantee coherence between the currency (Dollar) and the data, we focus specifically on the **United States of America (USA)**.

- **Source:** World Bank - Inflation, consumer prices (annual %)
- **Main variable:** Annual inflation rate in the USA (in percentage).

Data Preparation and Parameters

```
# Read data directly from Datahub.io
url <- "https://datahub.io/world-development-indicators/fp.cpi.totl.zg/_r/-/data.csv"
df_raw <- read.csv(url, stringsAsFactors = FALSE)

# Filter for the United States and clean columns
df_usa <- df_raw %>%
  filter(Country.Code == "USA") %>%
  select(Year, Inflation = Value) %>%
  filter(!is.na(Inflation)) %>%
  arrange(Year)

# Extract historical parameters
average_inflation <- mean(df_usa$Inflation) / 100
```

```r
inflation_sd <- sd(df_usa$Inflation) / 100

initial_savings <- 10000
num_years <- 10
num_simulations <- 10000

cat("Historical period:", min(df_usa$Year), "-", max(df_usa$Year), "\n")
```

```
## Historical period: 1960 - 2024
```

```r
cat("Number of observations:", nrow(df_usa), "\n")
```

```
## Number of observations: 65
```

```r
cat("Historical average inflation:", round(average_inflation * 100, 2), "%\n")
```

```
## Historical average inflation: 3.76 %
```

Methodology: Moving Block Bootstrap

A simple historical simulation (where individual years are sampled) assumes that the inflation of one year is completely independent of the previous year. In reality, economic crises cause periods of high inflation that last for several years (autocorrelation).

To solve this limitation, we use **Moving Block Bootstrap** with **3-year blocks**. This algorithm randomly samples groups of 3 consecutive years, thus preserving the actual market dynamics during crises.

```r
set.seed(123)

# Define block length
block_length <- 3
num_blocks <- ceiling(num_years / block_length)

# Function to generate a 10-year scenario
generate_scenario <- function() {
  start_indices <- sample(1:(nrow(df_usa) - block_length + 1),
                          num_blocks,
                          replace = TRUE)

  scenario <- unlist(lapply(start_indices, function(idx) {
    df_usa$Inflation[idx:(idx + block_length - 1)] / 100
  }))

  return(scenario[1:num_years])
}

# Run 10,000 simulations
simulated_inflation <- t(replicate(num_simulations, generate_scenario()))

# Calculate the real value of savings after 10 years
real_value <- initial_savings / apply(1 + simulated_inflation, 1, prod)
```

Analysis of Results and Risk

```r
# Expected mean value
mean_value <- mean(real_value)

# Probability of losing more than 20% (real value < 8000 $)
prob_loss_20 <- mean(real_value < 8000) * 100

# VaR (Value at Risk) 95% - financial standard
loss <- initial_savings - real_value
VaR_95_loss <- quantile(loss, 0.95)

cat("--- KEY RESULTS ---\n")
```

```
## --- KEY RESULTS ---
```

```r
cat("Expected mean value of savings:", round(mean_value, 2), "$\n")
```

```
## Expected mean value of savings: 6952.56 $
```

```r
cat("Probability of losing more than 20% of purchasing power:", round(prob_loss_20, 2), "%\n")
```

```
## Probability of losing more than 20% of purchasing power: 91.16 %
```

```r
cat("VaR (95%): The 95th percentile of losses is", round(VaR_95_loss, 2), "$\n")
```

```
## VaR (95%): The 95th percentile of losses is 4498.12 $
```

Sensitivity Analysis

The choice of 3-year blocks is a modeling assumption. To verify that our conclusions do not depend arbitrarily on this choice, we perform a sensitivity analysis by testing blocks of 2, 3, and 5 years.

```r
set.seed(456)
comparison <- data.frame()

for (L in c(2, 3, 5)) {
  n_blocks <- ceiling(num_years / L)

  generate_scenario_L <- function() {
    idx <- sample(1:(nrow(df_usa) - L + 1), n_blocks, replace = TRUE)
    scenario <- unlist(lapply(idx, function(i) df_usa$Inflation[i:(i + L - 1)] / 100))
    return(scenario[1:num_years])
  }

  simulations_L <- t(replicate(num_simulations, generate_scenario_L()))
  values_L <- initial_savings / apply(1 + simulations_L, 1, prod)
  losses_L <- initial_savings - values_L
```

```
  comparison <- rbind(comparison, data.frame(
    Block = paste(L, "years"),
    Mean = mean(values_L),
    Prob_Loss_20 = mean(values_L < 8000) * 100,
    VaR_95 = quantile(losses_L, 0.95)
  ))
}
comparison
```

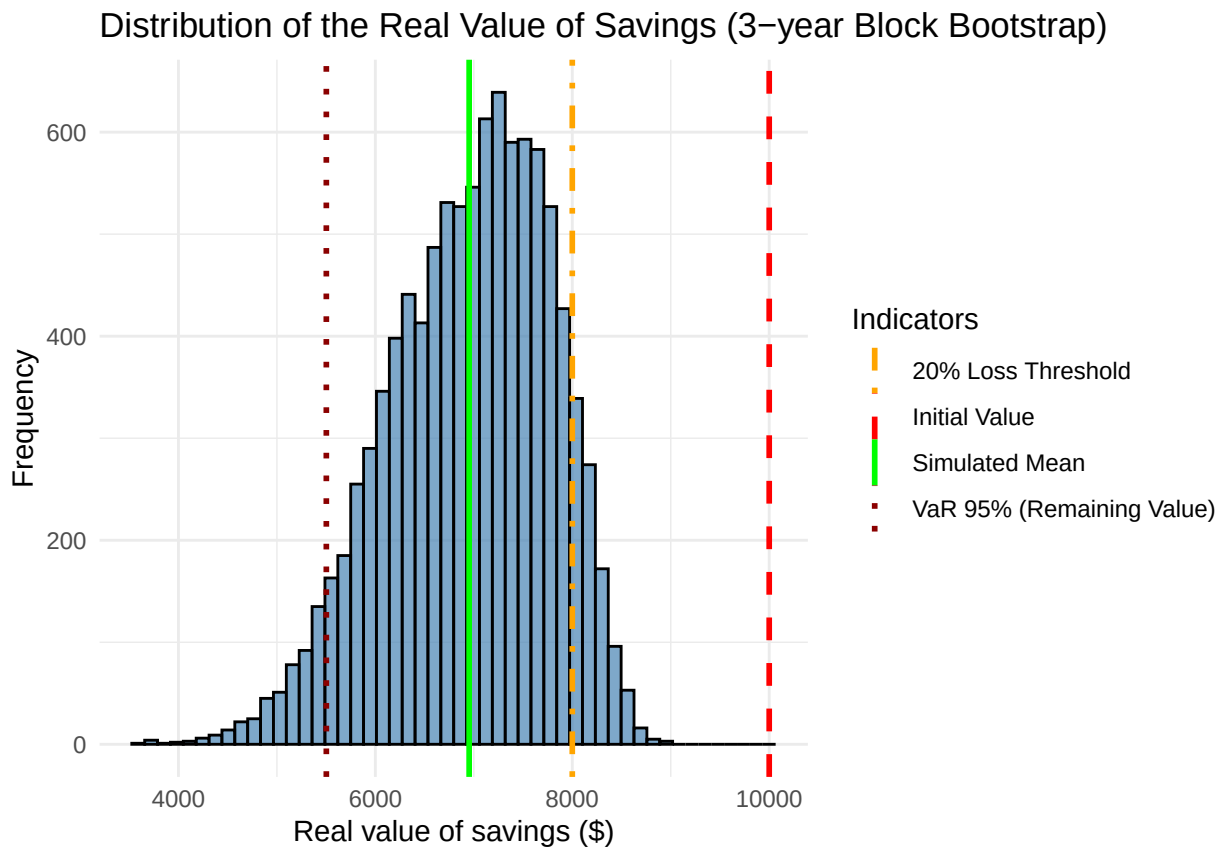
```
##        Block     Mean Prob_Loss_20   VaR_95
## 95%  2 years 6971.696        93.41 4322.931
## 95%1 3 years 6962.597        90.70 4496.969
## 95%2 5 years 6982.894        84.76 4750.125
```

Visualization

The histogram shows the distribution of the real value of savings for our main scenario (3-year blocks).

```
df_sim <- data.frame(Real_Value = real_value)

ggplot(df_sim, aes(x = Real_Value)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "black", alpha = 0.7) +
  geom_vline(aes(xintercept = 10000, color = "Initial Value"), linetype = "dashed", linewidth = 1) +
  geom_vline(aes(xintercept = mean_value, color = "Simulated Mean"), linewidth = 1) +
  geom_vline(aes(xintercept = 8000, color = "20% Loss Threshold"), linetype = "dotdash", linewidth = 1)
  geom_vline(aes(xintercept = 10000 - VaR_95_loss, color = "VaR 95% (Remaining Value)"), linetype = "do
  scale_color_manual(name = "Indicators",
                     values = c("Initial Value" = "red",
                                "Simulated Mean" = "green",
                                "20% Loss Threshold" = "orange",
                                "VaR 95% (Remaining Value)" = "darkred")) +
  labs(
    title = "Distribution of the Real Value of Savings (3-year Block Bootstrap)",
    x = "Real value of savings ($)",
    y = "Frequency"
  ) +
  theme_minimal()
```



Interpretation of Results

From the 10,000 simulated scenarios, we observe the following: 1. **Expected mean value** after 10 years is 6953 $. This indicates that the purchasing power of 10,000 $ will fall on average by 3047 $ under these historical conditions. 2. **The probability of losing more than 20%** of purchasing power is 91.2%. 3. **VaR (95%):** In the worst 5% of scenarios, the loss of purchasing power will be **above 4498 $**.

Sensitivity Analysis confirms that even if we change the length of the blocks (2 or 5 years), the underlying risk remains almost the same. This makes our model **robust** (stable) and shows that our conclusions do not depend on the arbitrary choice of a 3-year block.

Model Limitations

Although Block Bootstrap preserves short-term dependence, it has some limitations: 1. **Artificial transitions:** When a 3-year block ends, the next block starts from a completely random historical year. The transition between blocks does not reflect real economic continuity. 2. **Historical representation:** The model assumes that the future will behave like the past (1960-2024). If a new, structural crisis occurs (e.g., a global war), history does not include it.

Therefore, the results should be interpreted as possible scenarios based on the model's assumptions, and not as absolute predictions.

Conclusion

This simulation demonstrates the power of Monte Carlo Simulation with Block Bootstrap. By not limiting ourselves to a normality assumption and by preserving the historical memory of crises, the model clearly shows that holding savings in cash is a high-risk strategy for purchasing power erosion. To protect wealth, it is necessary to seek investment returns that exceed average inflation rates.